Practical Assignment: Smart Irrigation Node (ESP32)

Objective:

Develop embedded firmware on an ESP32 board that simulates a low-power smart irrigation node. The system should read environmental data (mocked or analog), control an actuator (LED), simulate LoRaWAN payload generation, and implement power-saving deep sleep cycles.

Requirements:

- ESP32 development board (required).
- Basic components: LED, resistors, wires (required).
- Potentiometer (optional – for analog soil moisture simulation).
- Button (optional – to simulate user input or wake-up source).

Assignment Tasks:

1. 1. Sensor Simulation: - Simulate soil moisture using a potentiometer or mocked analog values. - Convert readings to moisture percentage (0–100%).
2. 2. Actuator Control: - Use an LED to represent a solenoid valve. - Turn ON if moisture < 30%, OFF otherwise.
3. 3. LoRaWAN Packet Simulation: - Format and print payload (e.g., 0x1E for 30% moisture). - Simulate sending over LoRa by outputting to Serial Monitor.
4. 4. Power Optimization: - Use deep sleep mode for 1 minute between readings. - Wake using RTC timer and preserve state using RTC memory or EEPROM.

Deliverables:

- Firmware source code with comments.
- README explaining the working, payload format, and deep sleep strategy.
- Optional: Short video or screenshots of the system running (Serial output + LED behavior).

Evaluation Criteria:

- Correctness of sensor simulation and actuator logic.
- Proper use of deep sleep and wake-up functionality.
- Structure and clarity of code and documentation.
- Efficiency and thoughtfulness in payload construction.
- Bonus: EEPROM use, CRC in payload, threshold configuration, simulated HTTP/cloud interaction.

Timeframe:

Basic version: 4–6 hours. Full version with extras: 1 day max.